

ONA II, Norway

WMO: 012120

Lat: 62.8586N

Lon: 6.5378E

Elev: 15

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.6	-2.0	-10.4	1.5	-1.6	-8.6	1.8	-0.7	23.4	5.8	21.5	5.8	5.9	100	0.855

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.2	19.4	15.4	18.0	14.8	16.9	14.3	16.2	18.3	15.5	17.3	14.8	16.5	4.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.9	17.2	14.6	10.4	16.5	13.9	9.9	16.0	45.6	18.4	43.4	17.4	41.5	16.5	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.2	18.6	16.4	DB	-4.1	22.7	2.0	1.8	-5.6	23.9	-6.8	25.0	-7.9	26.0	-9.4	27.3	(4)
(5)			WB	-5.9	17.3	1.8	1.3	-7.1	18.2	-8.2	19.0	-9.2	19.7	-10.4	20.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	3.6	3.4	3.9	6.1	8.5	11.1	13.7	14.3	12.5	9.2	6.3	4.5	(6)	
	DBStd	4.56	2.53	2.55	2.39	2.30	2.39	2.06	2.13	1.81	2.15	2.65	2.81	2.94	(7)	
	HDD10.0	1094	198	186	188	119	58	9	0	0	4	47	114	170	(8)	
	HDD18.3	3729	457	419	447	367	304	217	146	126	174	283	362	428	(9)	
	CDD10.0	409	0	0	0	2	13	43	113	133	80	22	2	0	(10)	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)	
	CDH23.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.5	9.0	8.2	8.4	7.0	6.5	6.6	5.8	6.1	7.9	7.7	7.7	9.4	(14)	
(15) Precipitation	PrecAvg	1221	109	97	99	75	60	76	80	86	128	143	111	145	(15)	
	PrecMax	1630	229	206	323	155	101	202	161	214	283	258	233	245	(16)	
	PrecMin	445	0	0	0	16	7	1	19	1	38	18	0	15	(17)	
	PrecStd	244	76	67	74	44	26	47	35	47	70	68	74	59	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	9.2	9.7	15.0	17.4	19.7	22.3	21.3	18.8	16.6	12.8	10.9	(19)	
		MCWB	6.4	5.8	5.9	9.7	11.7	14.7	17.0	16.4	14.0	11.7	10.1	7.4	(20)	
	2%	DB	8.6	8.1	8.5	12.0	15.0	17.0	19.6	19.2	17.2	14.6	11.6	9.8	(21)	
		MCWB	6.0	5.5	5.6	8.1	10.8	13.6	16.0	15.6	13.5	11.3	9.2	7.4	(22)	
	5%	DB	7.7	7.3	7.7	10.2	13.1	15.2	17.9	17.8	16.2	13.4	10.8	9.1	(23)	
		MCWB	5.7	5.1	5.3	7.2	10.1	12.5	15.2	15.2	13.4	10.6	8.3	7.0	(24)	
	10%	DB	7.0	6.7	7.0	9.0	11.7	13.8	16.6	16.8	15.4	12.6	10.0	8.4	(25)	
		MCWB	5.3	4.7	5.0	6.7	9.3	11.7	14.5	14.7	13.0	10.1	7.9	6.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	7.2	7.2	10.1	12.4	15.3	17.5	17.6	15.4	12.7	10.9	9.0	(27)	
		MCDB	8.5	8.1	8.2	14.0	15.8	18.4	21.1	19.7	16.8	14.5	12.0	9.8	(28)	
	2%	WB	6.9	6.4	6.5	8.6	11.2	14.0	16.4	16.4	14.7	12.0	9.8	8.3	(29)	
		MCDB	7.9	7.3	7.6	11.0	14.2	16.7	18.9	18.2	16.0	13.9	11.1	9.4	(30)	
	5%	WB	6.1	5.7	5.7	7.8	10.4	12.8	15.5	15.7	14.0	11.2	8.9	7.5	(31)	
		MCDB	7.4	6.8	7.1	9.6	12.8	14.6	17.5	17.2	15.5	13.1	10.3	8.8	(32)	
	10%	WB	5.4	5.1	5.1	7.1	9.6	11.9	14.7	15.0	13.3	10.4	8.0	6.5	(33)	
		MCDB	6.8	6.4	6.6	8.6	11.5	13.5	16.6	16.5	15.0	12.4	9.8	8.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.8	2.7	3.1	3.6	3.8	3.2	3.4	3.2	3.1	2.9	2.8	2.9	(35)	
		MCDBR	3.3	3.1	3.5	5.5	6.0	5.4	5.6	4.6	3.8	3.7	3.2	3.8	(36)	
	5% WB	MCWBR	3.0	2.6	2.5	3.0	3.1	3.0	3.0	2.7	2.4	2.7	2.7	3.2	(37)	
		MCWBR	3.1	3.0	3.3	4.8	5.5	4.7	5.3	4.0	3.4	3.5	3.2	3.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.317	0.340	0.361	0.356	0.356	0.366	0.359	0.344	0.318	0.387	0.361	(40)	
		taud	2.083	2.256	2.329	2.367	2.443	2.479	2.477	2.496	2.513	2.417	2.664	2.464	(41)	
	Edn at Noon		364	619	758	823	861	867	847	821	762	626	358	192	(42)	
		Edh at Noon	33	61	82	97	98	96	94	85	70	52	30	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.14	0.60	1.71	3.33	4.61	4.93	4.59	3.52	2.06	0.92	0.23	0.08	(44)	
	RadStd		0.03	0.12	0.27	0.36	0.26	0.50	0.39	0.33	0.28	0.14	0.06	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		+0.50	+0.68	N/A	N/A	N/A	N/A	-150	-219	+63	N/A

Nomenclature: See separate page